MILITARY SPECIFICATION

LUBRICATING OIL, MACHINE TOOL SLIDEWAYS

1. SCOPE

1.1 Scope. This specification covers compounded lubricating oils for machine tool slideways (see 6.1).

1.2 Classification. The lubricating oils shall be of the following types (see 6.2).

Type I - Medium
Type II - Heavy
Type III - Heavy, special

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels and Related Products; Methods of Testing

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

ASTM Standards on Petroleum Products and Lubricants (Parts 17 and 18)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The lubricating oils shall consist of refined petroleum oils, with or without additives, as necessary.

3.2 Physical and chemical requirements. The lubricating oils shall conform to the requirements specified in table I and in 3.2.1.

Table I. Physical and chemical requirements (see 4.6)

Properties	Type I	Type II	Type III
Viscosity at 100°F. (37.78°C.)			
Kinematic, centistokes	61.0-75.0	195.0-238.0	195.0-238.0
(Saybolt, S.U.S.)1/	(282.9-347.6)	(903.8-1103.1)	(903.8-1103.1)
Flash point			
°F., minimum	300	300	300
(°C., minimum	(148.9)	(148.9)	(148.9)
Pour point			
°F., maximum	30	30	30
(°C., maximum	(-1.1)	(-1.1)	(-1.1
Copper corrosion; ASTM classification, maximum	3	3	3
Corrosiveness	No rusting, etching, or pitting of steel (Types I, II, and III)		
Ratio of coefficients of friction, static to kinetic (stick-slip ratio)	0.85	0.85	0.85

1/ For information only.

3.2.1 Vibratory load. Type III lubricating oil, when evaluated on vibratory load test equipment, shall have a minimum average value of 1800 pounds per square inch for four individual tests. The minimum value of any single test shall be 1200 pounds per square inch (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous mixture of lubricating oil offered for acceptance in a single isolated container, or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of drums or other unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of lubricating oil from a single isolated container; or filled with a homogeneous mixture of lubricating oil manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling for examination of the preparation for delivery. Take a random sample of packed containers from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) = 2.5 percent defective. Examine the sample in accordance with 4.4.1.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM method D 270. Test the samples in accordance with 4.6.

4.4 Inspection. Perform inspection in accordance with method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of the preparation for delivery. Examine samples taken in accordance with 4.3.1 for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests. All tests are quality conformance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

4.6 Test methods. Perform tests in accordance with the applicable methods listed in table II. Verification of the requirements for the ratio of the coefficients of static-to-kinetic friction (see 3.2; table I) and vibratory load (see 3.2.1) shall be accomplished through certification, as specified in 4.6.1.

Table II. Test methods

Test	Test method No., Fed. Std. 791	Test method No., ASTM
Viscosity, kinematic		D 445
Viscosity, conversion of kinematic to Saybolt		D 2161
Flash point		D 92
Pour point		D 97
Copper corrosion		D 130 1/
Corrosiveness (bimetallic couple)	5322	
Ratio of coefficients of friction, static to kinetic (stick-slip ratio)	By certification (see 4.6)	
Vibratory load	By certification (see 4.6)	

1/ Use a test temperature of 212°F. (100°C.) for a period of 3 hours ± 5 minutes.

4.6.1 Certification. Certification shall be accomplished by either one of the following two procedures.

(a) A photographic reproduction of a publication showing that a machine tool manufacturer normally engaged in the laboratory qualification of slideway lubricants has qualified the slideway lubricant offered by the supplier for acceptance. The photographic reproduction shall show the name of the machine tool manufacturer and the trade name of the lubricant offered for acceptance.

(b) A notarized statement by the lubricant supplier stating that the slideway lubricant offered for acceptance has been qualified by a machine tool manufacturer normally engaged in the laboratory qualification of slideway lubricants. The statement shall contain the name of the machine tool manufacturer, the trade name of the qualified slideway lubricant, and the date on which the qualification tests were made.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Unless otherwise specified in the contract or order (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290.

6. NOTES

6.1 Intended use.

Type I: This lubricating oil is intended for use on lightly to moderately loaded machine-tool slideways when oil in the viscosity range of 61 to 75 centistokes (282.9 - 347.6 S.U.S.) at 100°F. (37.78°C.) is specified by the tool manufacturer. It may also be used in other applications where a coefficient of static-to-kinetic friction ratio of less than 1.0 is required.

Type II: This lubricating oil is intended for use on heavily loaded machine-tool slideways when oil in the viscosity range of 195 to 238 centistokes (903.8 - 1103.1 S.U.S) at 100°F. (37.78°C.) is specified by the tool manufacturer. It may also be used in other applications where a heavy oil with a coefficient of static-to-kinetic friction ratio of less than 1.0 is required.

Type III: This lubricating oil is intended for use on heavily loaded machine-tool slideways when oil in the viscosity range of 195 to 238 centistokes (903.8 - 1103.1 S.U.S.) at 100°F. (37.78°C.) is specified by the tool manufacturer and on machine tools subject to vibrations which promote metal pickup. It may also be used on medium and highly loaded ferrous worm-wheels driven by a hardened steel worm.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of lubricating oil (see 1.2).
(c) Quantity of lubricating oil.
(d) Type and size of containers (see 5.1).
(e) Level of packaging and packing (see 5.1).

6.3 Storage conditions. The slideway lubricating oils covered by this specification may be stored at temperatures ranging from -70° to +120°F. (-57° to +49°C.).

Custodians:
Army - MR

Preparing activity:
Army - MR

Review activities
Army - ME, MU, WC

Project No. 9150-A037

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
U.S. Army Materials Research Agency
ATTN: AMXMR-TMS
Watertown, Massachusetts 02172

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification and how to insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-L-46017(M), Lubricating Oil, Machine Tool Slideways

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:40Z
Check the source to verify that this is the current version before use.